SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR.



Predictive Analytics

Human Activity Recognition Using Smartphone Sensor Data: A Comparative Predictive Modelling Study Using the KDD Process

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1. Problem Statement

1.1 Background and Motivation

Smartphones commonly contain accelerometers and gyroscopes that record linear acceleration and rotational movement. These measurements can be transformed into numerical features that describe a person's movement over short time windows. Human Activity Recognition (HAR) uses such data to classify activities such as walking, sitting, standing, and lying down.

Reliable activity recognition can support context-aware mobile applications, fitness tracking, assisted living research, and human-computer interaction. However, some activities produce similar sensor patterns. Sitting and standing may be confused, and postural transitions can be particularly difficult to identify when only a small number of examples are available.

1.2 Problem Definition

The project investigates how effectively supervised machine-learning models can classify human activities from smartphone sensor-derived features. It compares model performance on the UCI Human Activity Recognition Using Smartphones (HAR) dataset and the UCI Smartphone-Based Recognition of Human Activities and Postural Transitions (HAPT) dataset. The study also examines the effect of additional transition classes and class imbalance on classification performance.

1.3 Objectives

- To understand and inspect the UCI HAR and UCI HAPT datasets, including their feature dimensions, activity labels, and class distributions.
- To perform data quality checks and prepare the feature matrices and activity labels for modelling.
- To implement and compare Logistic Regression, k-Nearest Neighbors (k-NN), Decision Tree, and Random Forest.
- To select the strongest model using validation performance, particularly macro F1-score.
- To evaluate the selected model on the official test partitions using accuracy, precision, recall, macro F1-score, and confusion matrices.
- To interpret class-level errors and Random Forest feature importance, and discuss limitations and future improvements.

1.4 Scope of the Project

The project uses the pre-engineered 561-feature representations supplied by the two UCI datasets. UCI HAR contains six basic activity classes; HAPT includes the same six activities and six additional postural transitions. Four supervised classifiers are compared. The project

does not collect new sensor recordings, use raw-signal deep learning, or claim that the two related datasets constitute independent external validation.

1.5 Significance

The project provides a reproducible comparison of conventional classification methods on two closely related activity-recognition datasets. It demonstrates why accuracy alone may be insufficient when some activity classes are rare and why macro F1-score and confusion matrices are useful for a more balanced assessment.

2. Dataset

2.1 Source and Description

Both datasets are available through the UCI Machine Learning Repository. They contain features extracted from smartphone accelerometer and gyroscope signals collected during human activities.

- **UCI HAR:** https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones
- **UCI HAPT:** https://archive.ics.uci.edu/dataset/341/smartphone+based+recognition+of+human+activities+and+postural+transitions

2.2 Attribute Description

Attribute group	Description
Sensor-derived input features	561 engineered time- and frequency-domain features from smartphone motion signals.
Target label	Integer-coded activity class, mapped to a human-readable activity name.
HAR classes	LAYING, SITTING, STANDING, WALKING, WALKING_DOWNSTAIRS, WALKING_UPSTAIRS.
HAPT classes	The six HAR activities plus STAND_TO_LIE, SIT_TO_LIE, LIE_TO_SIT, LIE_TO_STAND, STAND_TO_SIT, SIT_TO_STAND.

2.3 Dataset Summary

Property	UCI HAR	UCI HAPT
Training samples	7,352	7,767
Test samples	2,947	3,162
Input features	561	561
Number of classes	6	12
Missing feature values	0 reported	0 reported
Duplicate training feature rows	0 found in quality audit	0 found in quality audit

2.4 Characteristics and Limitations

The features are already engineered and scaled approximately between −1 and 1 in the supplied datasets. HAR focuses on common activities, while HAPT adds transition labels with much smaller training counts. This difference makes HAPT a more challenging classification setting and affects the interpretation of macro-averaged metrics.

- The datasets are related and should not be described as independent external validation.
- The project uses engineered features rather than raw sensor time series.
- The small number of examples for some transition classes can limit generalisation.

3. Data Understanding and Preprocessing

3.1 Data Understanding

The first stage examined the dimensions of each dataset, feature count, target labels, missing values, duplicate training feature rows, and activity-class distributions. These checks confirmed that both datasets were structured numerical feature tables suitable for supervised multiclass classification.

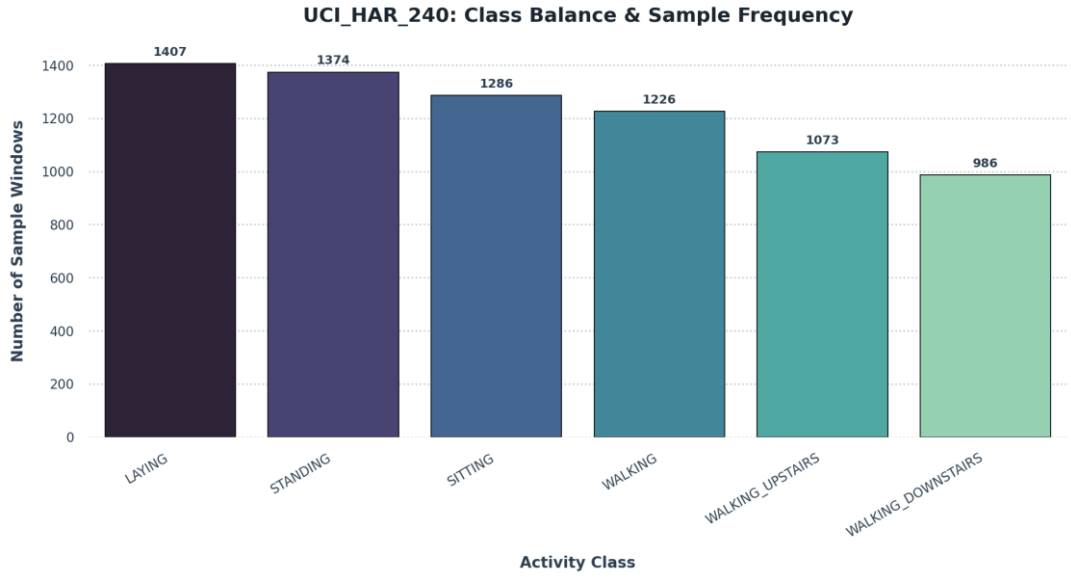


Figure 1. Training class distribution for UCI HAR.

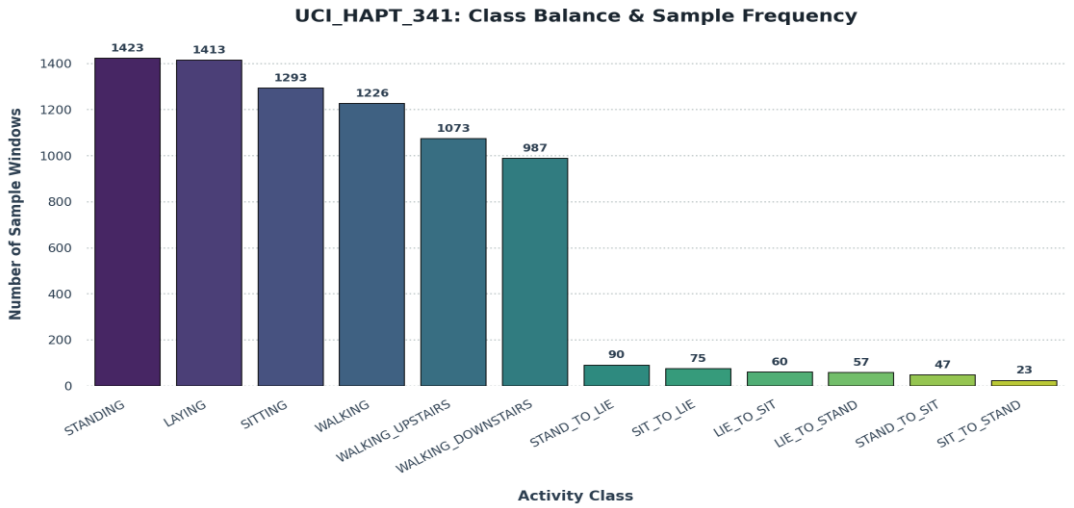


Figure 2. Training class distribution for UCI HAPT.

3.2 Data Quality Checks

The notebook's quality audit found zero missing feature cells in the training and test feature matrices and zero duplicate training feature rows. Since the source datasets are already prepared feature tables, extensive raw-data cleaning was not required.

3.3 Feature and Label Separation

For each dataset, the 561 sensor-derived columns were used as input features and the activity identifier was used as the target. The labels were retained as multiclass targets so the classifiers could learn to distinguish the six HAR classes and the twelve HAPT classes.

3.4 Validation Split and Scaling

The official training and test partitions were preserved. A stratified validation subset representing 20% of the training data was used for model comparison, with random state 42 as recorded in the presentation/notebook workflow. Scaling was applied where required by the model, especially for Logistic Regression and k-NN. The official test set was reserved for final evaluation after model selection.

3.5 Summary of Preprocessing

Step	Action	Purpose
1	Checked missing values and duplicate training feature rows.	Confirm basic data quality.
2	Separated input features and activity labels.	Create X and y for supervised learning.
3	Preserved official train/test partitions.	Keep final testing separate from model selection.
4	Created a stratified validation split from training data.	Compare models on a consistent validation set.
5	Applied scaling where required by the model.	Support scale-sensitive classifiers.

4. Predictive Modelling Approach

4.1 KDD Process

The project follows the Knowledge Discovery in Databases (KDD) process: data selection, data understanding, preprocessing, transformation, data mining, evaluation, and knowledge presentation.

1. **Data selection:** choose the UCI HAR and UCI HAPT datasets.
2. **Data understanding:** inspect feature dimensions, labels, data quality, and class distributions.
3. **Preprocessing and transformation:** separate features and labels, create validation data, and apply scaling where needed.
4. **Data mining:** train four supervised classifiers.
5. **Evaluation:** compare validation metrics and evaluate the selected model on the official test partition.
6. **Knowledge presentation:** report metric tables, confusion matrices, and feature-importance plots.

4.2 Models Used

Model	Principle	Role in comparison
Logistic Regression (LR)	Learns linear decision boundaries for multiclass classification.	Baseline model; scaling applied where needed.
k-Nearest Neighbors (k-NN)	Predicts a class from nearby examples in feature space.	Distance-based comparison model; k = 5 as recorded.
Decision Tree (DT)	Learns feature-based decision rules by recursively splitting data.	Nonlinear, interpretable tree-based model.
Random Forest (RF)	Combines predictions from an ensemble of decision trees.	Selected model on both datasets based on validation macro F1.

4.3 Evaluation Metrics

- **Accuracy:** proportion of all test examples classified correctly.
- **Precision:** proportion of predicted examples for a class that are correct; reported as macro average across classes.
- **Recall:** proportion of actual examples for a class correctly identified; reported as macro average across classes.
- **Macro F1-score:** unweighted mean of class-wise F1-scores, so each class contributes equally regardless of its frequency.
- **Confusion matrix:** counts correct predictions and shows which activity classes are confused with one another.

4.4 Model Selection Strategy

Models were compared on the validation subset. Validation macro F1-score was emphasized because it gives equal weight to each activity class, which is important for HAPT where transition classes are rare. Random Forest achieved the highest validation macro F1-score on both datasets and was selected for final testing.

4.5 Overall Methodology

The workflow is: load dataset → audit and understand data → separate features and labels → make stratified validation split → train four models → compare validation metrics → select Random Forest → retrain on the full training partition → evaluate once on the official test partition → interpret errors and feature importance.

5. Model Development

5.1 Tools and Environment

The implementation was carried out in a Python notebook. The workflow uses common data-science and machine-learning libraries, including pandas/NumPy for tabular data operations, scikit-learn for model training and evaluation, and Matplotlib for visualisation. The exact installed package versions should be recorded from the notebook environment if required for reproducibility.

5.2 Logistic Regression

Logistic Regression was used as a baseline classifier. It estimates class probabilities from a weighted combination of input features and assigns the most likely activity class. Because it is sensitive to feature scales, the workflow applies scaling where required.

5.3 k-Nearest Neighbors

k-NN classifies an example by examining nearby training examples in feature space. The project settings record k = 5 and distance-weighted voting. Scaling is important for distance-based methods so that feature magnitudes do not dominate similarity calculations.

5.4 Decision Tree

The Decision Tree model learns a sequence of feature-based splits to separate activity classes. It can capture nonlinear patterns, although a single tree may be less stable than an ensemble.

5.5 Random Forest

Random Forest combines multiple decision trees trained with randomness in samples and features. The notebook/presentation records 250 trees for the selected configuration. It was selected on both datasets because it achieved the highest validation macro F1-score. The trained forest also provides feature-importance scores for interpretation.

5.6 Implementation Workflow

1. Load the official training and test feature matrices and activity labels.
2. Inspect shapes, class frequencies, missing values, and duplicate feature rows.
3. Create a stratified validation split from the training partition.
4. Fit Logistic Regression, k-NN, Decision Tree, and Random Forest.
5. Compare validation accuracy and macro F1-score.
6. Select Random Forest and retrain it using the full official training partition.
7. Evaluate the selected model on the official test partition and generate plots.

6. Evaluation of the Model

6.1 Evaluation Strategy

Validation results were used to select the model, while the official test partitions were used for final performance reporting. The distinction matters because validation scores guide model choice, whereas test scores estimate performance on held-out examples.

6.2 Validation Comparison

Model	HAR accuracy	HAR macro F1	HAPT accuracy	HAPT macro F1
Logistic Regression	98.57%	98.68%	97.04%	84.02%
k-NN	96.47%	96.67%	94.92%	82.42%
Decision Tree	93.88%	93.80%	89.64%	79.06%
Random Forest	98.78%	98.79%	97.23%	86.61%

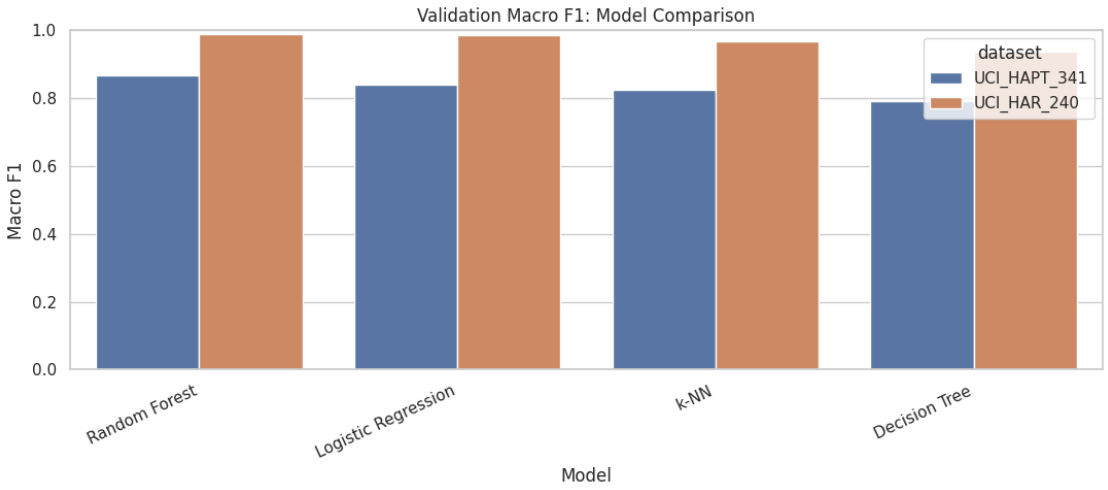


Figure 3. Validation macro F1-score comparison across the four models.

Random Forest achieved the highest validation macro F1-score for both HAR (98.79%) and HAPT (86.61%). The validation results therefore support selecting Random Forest for the final test evaluation.

6.3 Final Test Results

Metric	UCI HAR (Random Forest)	UCI HAPT (Random Forest)
Accuracy	92.84%	91.14%
Macro precision	92.94%	82.64%
Macro recall	92.54%	82.53%
Macro F1-score	92.66%	82.25%
Test samples	2,947	3,162

6.4 Confusion Matrix Analysis

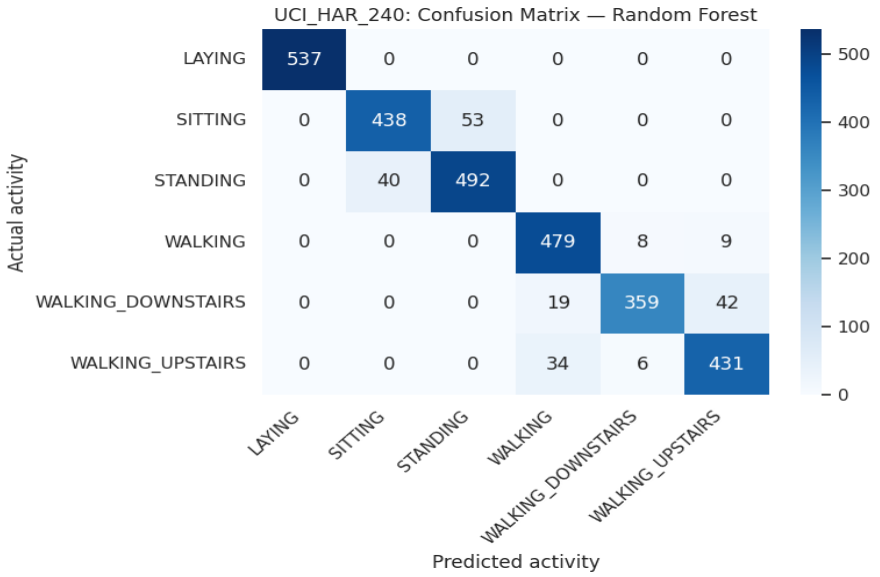


Figure 4. UCI HAR Random Forest confusion matrix on the official test set.

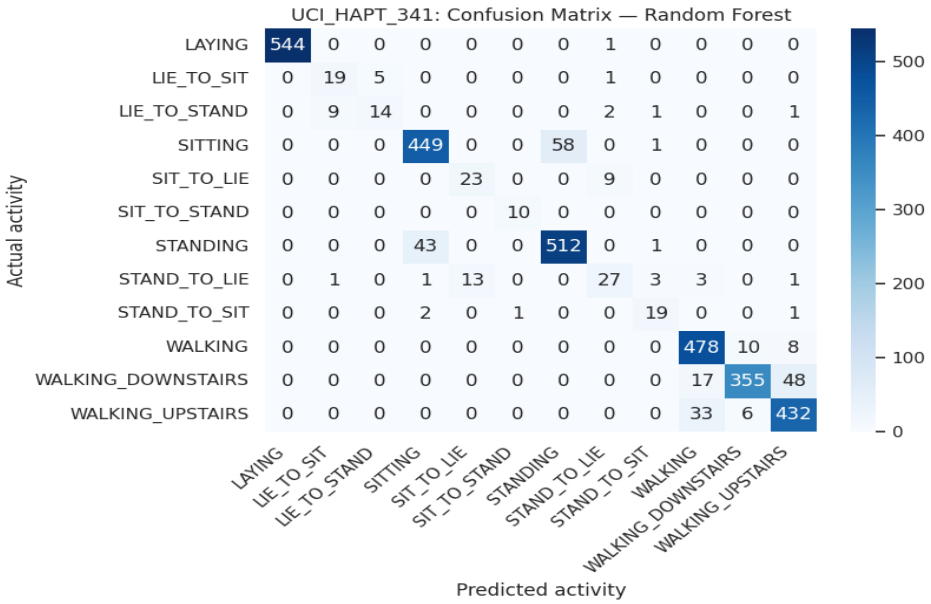


Figure 5. UCI HAPT Random Forest confusion matrix on the official test set.

For HAR, the model performs strongly on the basic activities, but some sitting examples are predicted as standing, reflecting the similarity of their sensor patterns. For HAPT, most errors are concentrated in rare transition classes. This pattern helps explain why HAPT macro F1-score (82.25%) is lower than its accuracy (91.14%).

6.5 Feature Importance

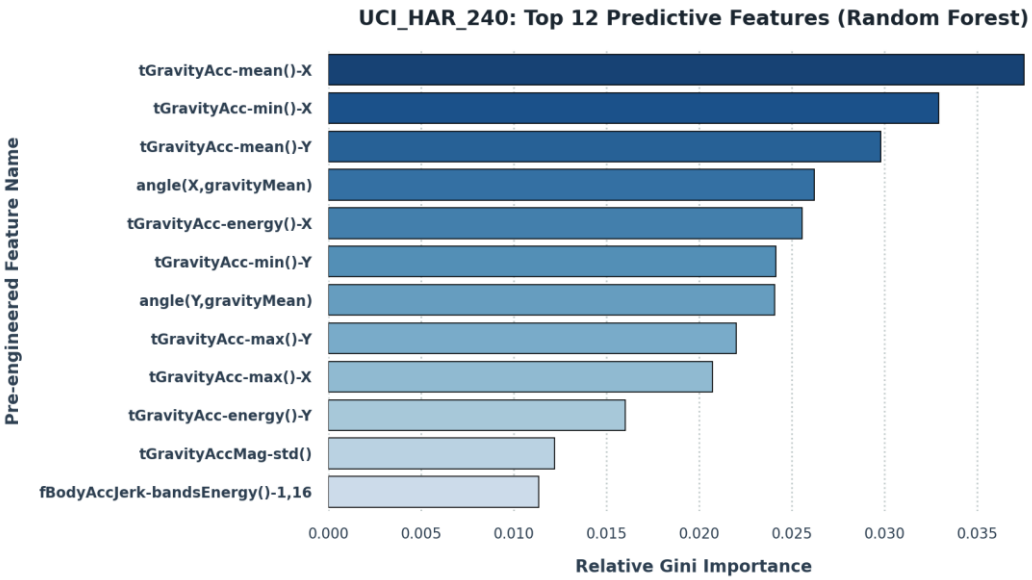


Figure 6. Random Forest feature importance for UCI HAR.

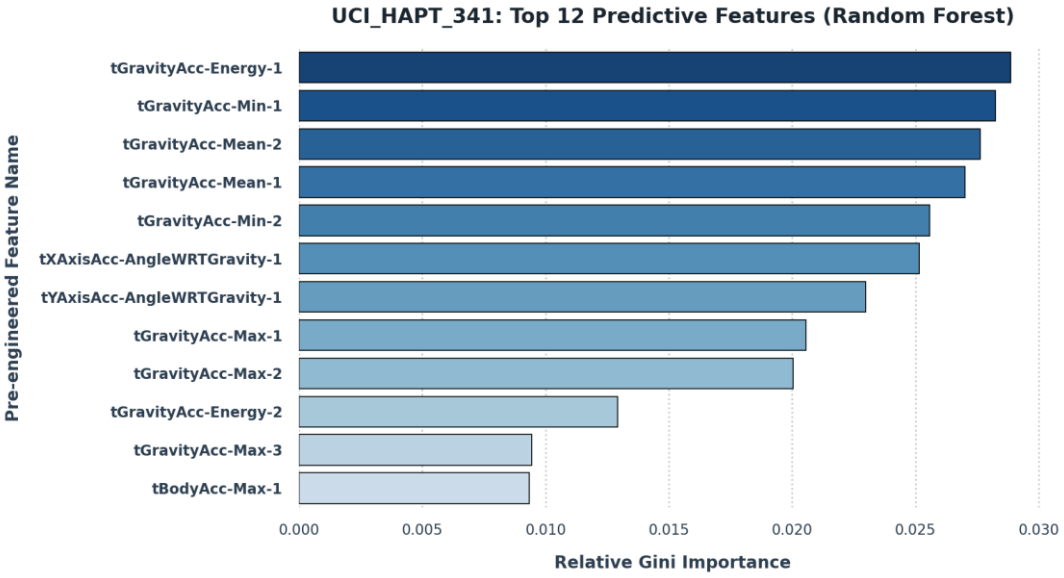


Figure 7. Random Forest feature importance for UCI HAPT.

The feature-importance plots rank the input features according to their contribution to Random Forest splits. Gravity-related acceleration features rank highly in the HAR analysis. Feature importance indicates predictive usefulness within the trained model; it does not establish a causal relationship.

7. Results and Interpretation

7.1 UCI HAR Results

On the official HAR test set, Random Forest achieved 92.84% accuracy and 92.66% macro F1-score. These close values indicate that performance is comparatively balanced across the six classes. LAYING was classified very reliably, while sitting and standing showed some confusion.

7.2 UCI HAPT Results

On the official HAPT test set, Random Forest achieved 91.14% accuracy and 82.25% macro F1-score. The gap between accuracy and macro F1 indicates that overall accuracy is supported by common classes while the rarer transition classes are harder to classify. The class-wise results in the notebook show F1-scores around 0.61 for LIE_TO_STAND and STAND_TO_LIE.

7.3 Comparison and Discussion

Observation	Interpretation
Random Forest led validation macro F1 on both datasets.	It provided the strongest balanced validation performance among the four models tested.
HAR test macro F1 was 92.66%.	The six basic activities were recognised comparatively consistently.
HAPT test macro F1 was 82.25%, below 91.14% accuracy.	Rare transition classes have fewer examples and lower class-wise performance.
Confusion matrices show activity-specific errors.	Aggregate metrics should be read alongside class-level results.
Feature importance highlights useful sensor-derived variables.	These rankings support interpretation but do not imply causation.

7.4 Practical Applications

Activity recognition from smartphone sensors can support fitness and activity logging, context-aware mobile applications, and research prototypes for monitoring daily movement. Any health-related or safety-critical use would require further validation on representative users and real-world recordings before deployment.

8. Conclusion

8.1 Summary of the Work

This project followed the KDD process to compare four supervised machine-learning classifiers on the UCI HAR and UCI HAPT datasets. Data quality checks, feature/label

separation, validation splitting, model comparison, final testing, confusion-matrix analysis, and feature-importance interpretation were completed.

8.2 Key Findings

- Random Forest achieved the highest validation macro F1-score on both datasets: 98.79% for HAR and 86.61% for HAPT.
- On the official test sets, Random Forest achieved 92.84% accuracy and 92.66% macro F1 on HAR.
- On HAPT, it achieved 91.14% accuracy and 82.25% macro F1, showing the additional challenge of recognising rare transitions.
- Confusion matrices and feature-importance plots provided additional insight beyond aggregate accuracy.

8.3 Limitations

- The study uses engineered features supplied by the UCI datasets rather than raw sensor sequences.
- HAR and HAPT are related datasets, not independent external validation sources.
- Rare HAPT transitions have limited training examples, which can reduce class-wise reliability.
- The reported results depend on the supplied datasets and chosen model settings; real-world deployment may perform differently.

8.4 Future Scope

- Investigate class weighting or carefully evaluated resampling methods for rare transition classes.
- Correct and validate an MLP pipeline before including neural networks in a fair comparison.
- Evaluate raw-signal CNN or LSTM models if raw sensor sequences are available.
- Use subject-wise validation and additional independent datasets to better assess generalisation.
- Add explainability methods and evaluate the model on realistic smartphone placements and user populations.

References

[1] Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). A public domain dataset for human activity recognition using smartphones. *Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN).*

[2] Reyes-Ortiz, J. L., Oneto, L., Samà, A., Parra, X., & Anguita, D. (2016). Transition-aware human activity recognition using smartphones. *Neurocomputing, 171*, 754–767. https://doi.org/10.1016/j.neucom.2015.07.085

[3] UCI Machine Learning Repository. *Human Activity Recognition Using Smartphones.* https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones

[4] UCI Machine Learning Repository. *Smartphone-Based Recognition of Human Activities and Postural Transitions.* https://archive.ics.uci.edu/dataset/341/smartphone+based+recognition+of+human+activities+and+postural+transitions

[5] scikit-learn developers. *scikit-learn: Machine Learning in Python.* https://scikit-learn.org/